

Example: The space $(C[0, 1], \|\cdot\|_\infty)$ is connected.

path-connected

$$f, g \in C[0, 1] \quad \exists F: [0, 1] \rightarrow C[0, 1]$$

① F - continuous

$$\textcircled{2} \quad F(0) = f \quad \& \quad F(1) = g$$

$$\mapsto \textcircled{3} \quad F(t) \in C[0, 1] \quad \forall t \in [0, 1]$$

the straight line segment
works



$$\underline{F(t)} = (1-t)f + t \cdot g \quad \text{Linear map}$$

$$F(0) = f \quad F(1) = g$$

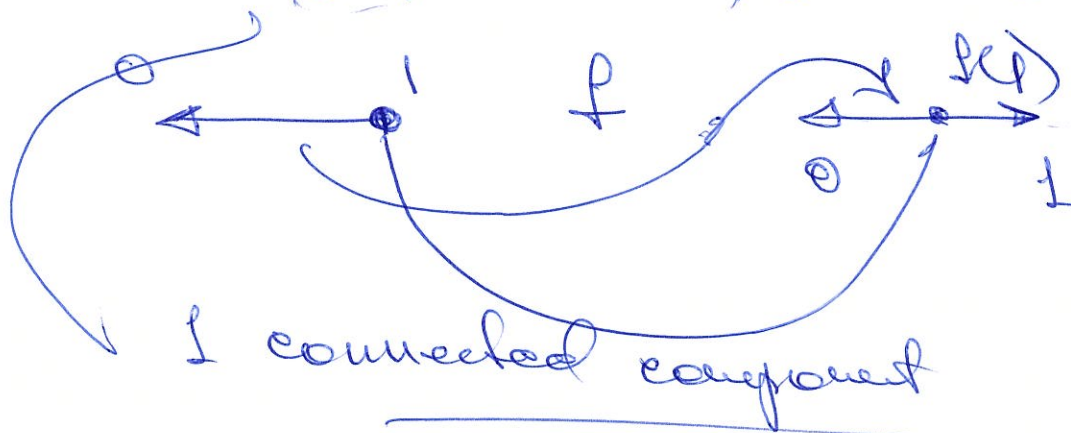
③ $\Leftarrow C[0, 1]$ - vector space

Example: The spaces $(0, 1)$ and $(0, 1]$ are not homeomorphic.

similar idea to page 113

Assume how: $f: [0, 1] \rightarrow (0, 1)$

$f: (0, 1) \rightarrow (0, 1) \setminus f(1) \leftarrow 2$ connected components



Question: Let (X, τ_X) and (Y, τ_Y) be topological spaces. How can we define a topology on the Cartesian product $X \times Y$?

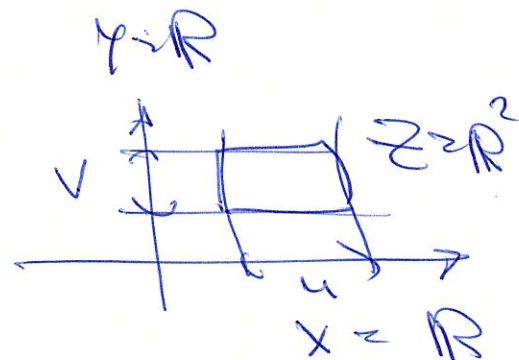
$$Z = \underline{X \times Y} = \{(x, y) : x \in X, y \in Y\}$$

Definition

Let (X, τ_X) and (Y, τ_Y) be topological spaces. The product topology on $\mathbb{Z} = X \times Y$ is the topology generated by the base

base of open rectangles $\rightarrow \{U \times V\}_{U \in \tau_X, V \in \tau_Y} \leftarrow$

① $\mathbb{Z} = \bigcup U_i \times V_i = X \times Y$
 $U_i \in \tau_X$ $V_i \in \tau_Y$



② $\forall W_i \in \mathcal{B} \quad i=1,2 \quad \exists W_3 \in \mathcal{B} \quad W_3 \subseteq W_1 \cap W_2$

$W_1 = U_1 \times V_1 \quad i=1,2 \quad W_3 = (U_1 \cap U_2) \times (V_1 \cap V_2)$
 $\xrightarrow{\quad}$

Lemma. $(x_n, y_n) \in \mathbb{Z}$

$(x_n, y_n) \rightarrow (x, y) \text{ in } \tau_{pr} \iff$

$\iff x_n \xrightarrow{\tau_X} x \quad \& \quad y_n \xrightarrow{\tau_Y} y$

Exercise: Show that the product topology on $\mathbb{R} \times \mathbb{R}$ (where each copy of $\mathbb{R}$ has the standard topology) is the standard topology on $\mathbb{R}^2$.

$$\tau_{\text{pr}, \mathbb{R}^2} = \tau_{\mathbb{R}^2} \text{ via } d_2\text{-metric}$$

$$\forall W \in \tau_{\text{pr}, \mathbb{R}^2} \Rightarrow W = \bigcup_{i \in I} U_i \times V_i$$

$U_i \in \tau_{\mathbb{R}}, V_i \in \tau_{\mathbb{R}}$

W - open in d_2 -metric

$$\text{~~to show~~ } (x, y) \in W \Rightarrow \exists \varepsilon > 0 \ B_{d_2}((x, y), \varepsilon) \subseteq W$$

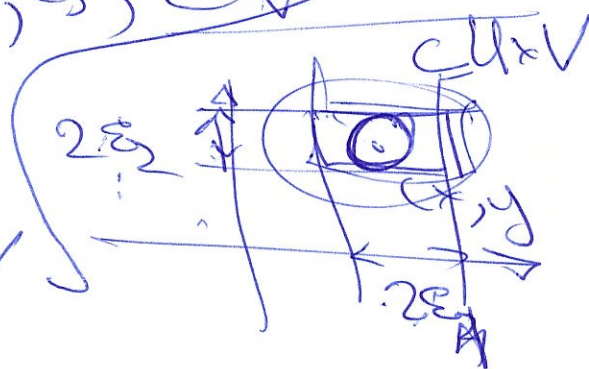
$$\exists U \in \tau_x, V \in \tau_y$$

$$x \in U \Rightarrow \exists \varepsilon_1 > 0 \ B(x, \varepsilon_1) \subseteq U$$

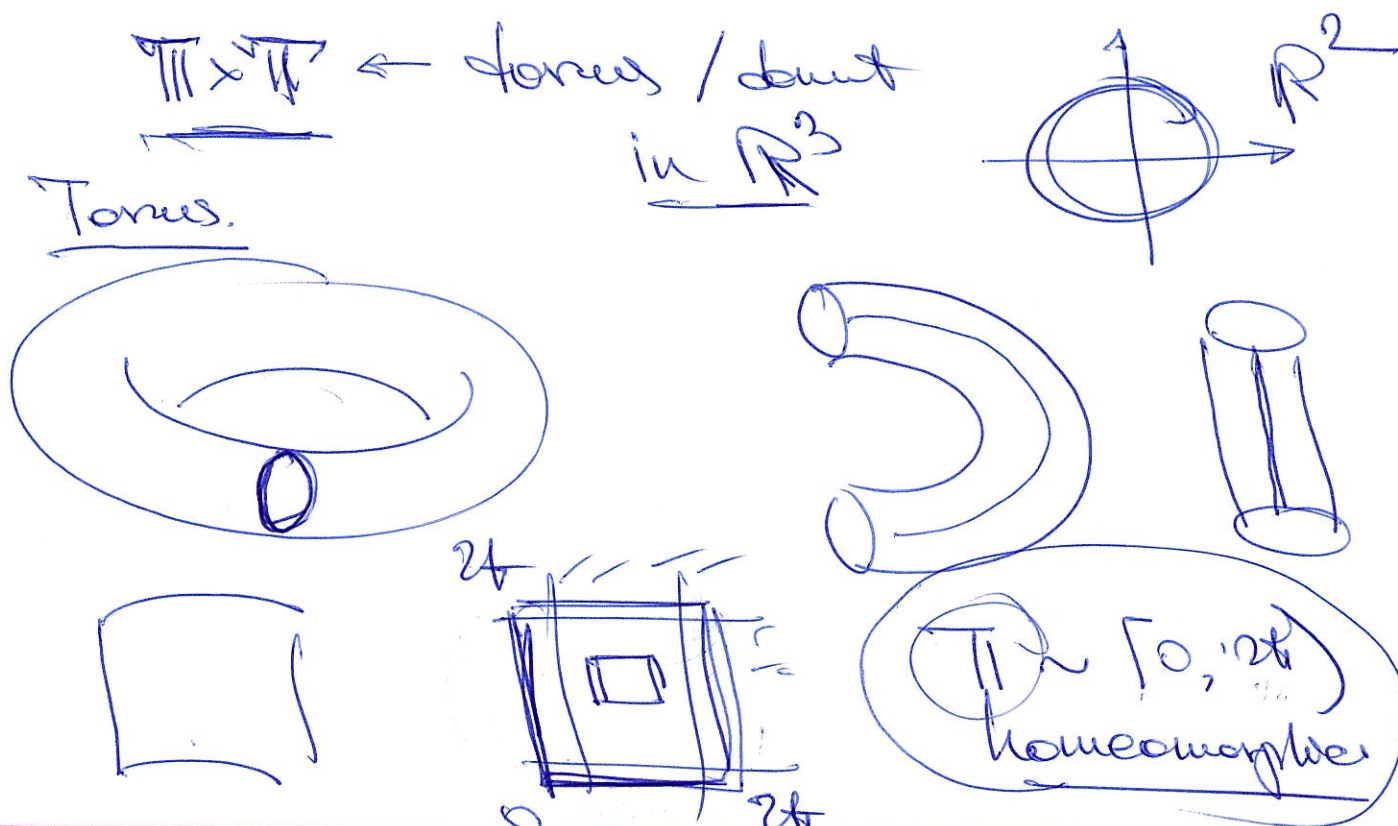
$$y \in V \Rightarrow \exists \varepsilon_2 > 0 \ B(y, \varepsilon_2) \subseteq V$$

$$\underline{B((x, y), \min(\varepsilon_1, \varepsilon_2))}$$

$$\subseteq U \times V \subseteq W$$

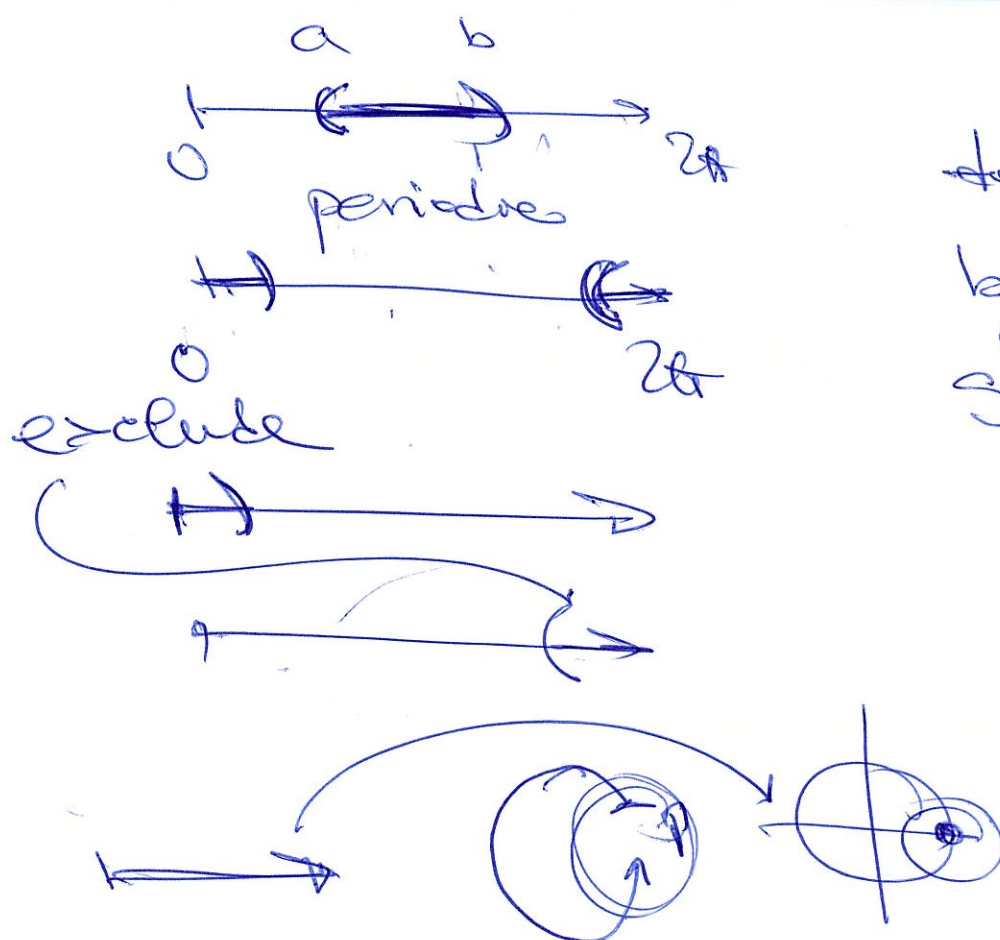


Exercise: Let $\mathbb{T}$ be the unit circle in the complex plane, with the standard topology. Describe the product topology on $\mathbb{T} \times \mathbb{T}$.



MATH3611 / MATH5705

126 / 133



$[0, 2\pi)$ with topology spanned by the subbases

$$S = \{ (a, b) \mid 0 \leq a < b < 2\pi \}$$

$$\{ [0, a) \cup (b, 2\pi) \mid 0 \leq a < b < 2\pi \}$$

Exercise: Let (X, τ_X) and (Y, τ_Y) be topological spaces, and suppose that $\mathcal{B}_X$ and $\mathcal{B}_Y$ are bases for τ_X and τ_Y , respectively. Then $\{U \times V\}_{U \in \mathcal{B}_X, V \in \mathcal{B}_Y}$ is a base for the product topology on $X \times Y$.

compare to page 124 and
adjust the solutions

Question: What about an arbitrary (possibly infinite) collection of topological spaces $\{(X_i, \tau_i)\}_{i \in I}$? Can we similarly put a topology on the Cartesian product $\prod_{i \in I} X_i$?

$$\mathcal{B} = \left\{ \prod_{i \in I} U_i \text{ where } U_i \in \tau_{X_i} \right\}$$

~~product~~ topology

$$U_i \in \tau_{X_i}$$

such approach is too restrictive!
 as the Base $\mathcal{B}$ is too nuch
 so $\tau(\mathcal{B})$ is too low!

Definition

Let $\{(X_i, \tau_i)\}_{i \in I}$ be a collection of topological spaces. The box topology $\prod_{i \in I} X_i$ is the topology with base

$$\mathcal{B} = \left\{ \prod_{i \in I} U_i : U_i \in \tau_i, \forall i \in I \right\}$$

Example: Consider $\mathbb{R}^{\mathbb{N}}$ (infinite Cartesian power of $\mathbb{R}$, i.e. the set of real sequences), with the box topology coming from the standard topology on each copy of $\mathbb{R}$. Show that the sequence

$$(1, 0, 0, \dots), (0, \frac{1}{2}, 0, \dots), (0, 0, \frac{1}{3}, \dots), \dots$$

does **not** converge to the zero sequence.

$\mathbb{R}^{\mathbb{N}}$ with box topology
 set of all real sequences
 $(x_n) \in \mathbb{R}^{\mathbb{N}}$
 $x_n = (x_n^{(1)}, x_n^{(2)}, \dots)$

$$x_n^{(k)} = \begin{cases} 0 & n \neq k \\ \frac{1}{n} & n = k \end{cases}$$

$$x_n \rightarrow \underline{0} = (0, 0, \dots)$$

$\forall U \in \tau_{\text{box}} \ni \underline{0}$ $\exists K: \forall n \geq K \ x_n \notin U$

$$U_0 = \prod_{i \in \mathbb{N}} \left(-\frac{1}{i+1}, \frac{1}{i+1} \right) \in \tau_{\text{Box}}$$

$\underline{0} \notin U_0$ so (by limit def) $\exists K:$

$\forall n \geq K \ x_n \notin U_0 \leftarrow \text{false if } n \neq k$

$$x_n \notin U_0 \iff \exists k \left(-\frac{1}{k+1}, \frac{1}{k+1} \right) \ni x_n^{(k)} \iff \frac{1}{n} \notin \left(-\frac{1}{k+1}, \frac{1}{k+1} \right)$$

Definition

Let $\{(X_i, \tau_i)\}_{i \in I}$ be a set of topological spaces. The **product topology** on

$X = \prod_{i \in I} X_i$ is the topology generated by the base

$$\mathcal{B} = \left\{ \prod_{i \in I} U_i : U_i \in \tau_i, \forall i \in I \text{ \& } U_i = X_i \text{ for all but finitely many } i \in I \right\}.$$

Lemma on page 124 applies to this case also

$$x_n \in X \quad x_n = (x_n^{(1)}, x_n^{(2)}, \dots)$$

$$\text{Assume } x_n \rightarrow x = (x_1, x_2, \dots)$$

$$\textcircled{*} \quad \forall U \in \mathcal{B}_{\text{pr}} \quad \exists K : \forall n \geq K \quad x_n \in U$$

we show, K fixed, then

$$\textcircled{*} \quad \forall U \in \mathcal{B}_K \quad \exists K \text{ s.t. } \forall n \geq K \quad x_n \in U$$

$$\text{For } U \in \mathcal{B}_K. \quad U \in \mathcal{B}_{\text{pr}} \quad U = \prod_{i \in I} V_i$$

$$V_i = \begin{cases} X_i & i \neq k \\ U & i = k \end{cases}$$

$$\text{Apply } \textcircled{*} \text{ to } U \quad \exists K : \forall n \geq K \quad x_n \in U \Rightarrow \exists K \text{ s.t. } x_n \in U$$

Exercise: Show that a sequence in $\mathbb{R}^N$ with the product topology converges iff it converges pointwise.

Lemmas from previous page
(also p. 124)

$$\mathbb{R}^X \rightarrow F(\mathbb{R}, X)$$

Exercise: More generally, let X be a set. Show that the product topology on $\mathbb{R}^X = F(X, \mathbb{R})$ is the topology of pointwise convergence.

ok Base of pr. tp.

$$B = \left\{ \prod_{i \in I} U_i : U_i = X_i \text{ except for finitely many } i \right\}$$

$$S = \left\{ \prod_{i \in I} U_i : U_i = \begin{cases} X_i & \text{if } i \neq k \\ U \subset X_k & \text{for one fixed } k \end{cases} \right\}$$

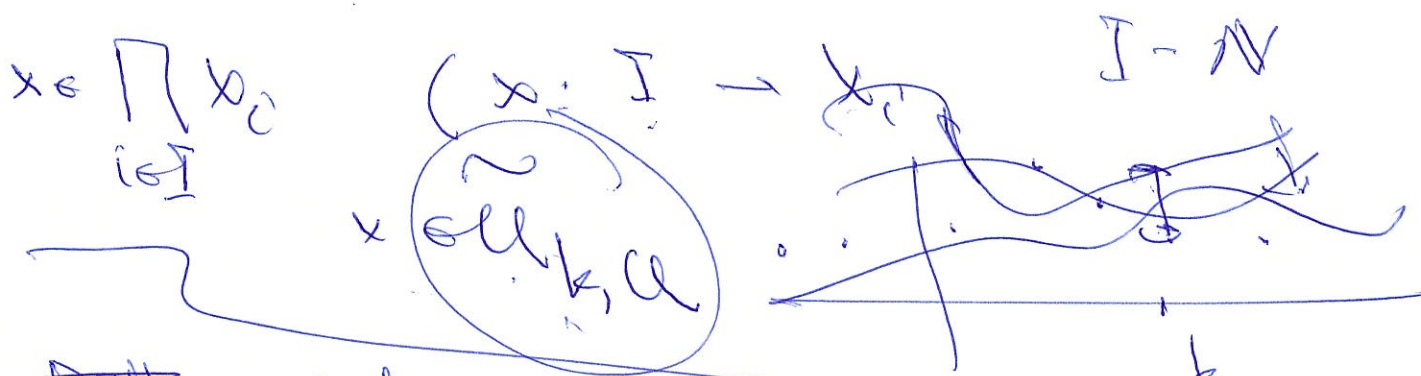
opense tll from page 131

MATH3611 / MATH5705

133 / 133

S -subbase for product topology

$$B = \left\{ \bigcap_{i=1}^m U_{k_i} : U_{k_i} \in S \right\} \quad X_i = \mathbb{R}$$



pointwise

convergence & topology is an example of product topology

gate sets in p.t.

$$\mathbb{R}^X = \prod_{x \in X} \mathbb{R} \equiv \{f: X \rightarrow \mathbb{R}\}$$

$$\prod_{i \in I} X_i = \{f: I \rightarrow \bigcup_{i \in I} X_i\}$$

cartesian product $\prod_{i \in I} X_i$ is
the set of maps / functions

$$f: I \rightarrow \bigcup_{i \in I} X_i \text{ s.t.}$$

$$f(i) \in X_i \quad \forall i \in I.$$